ALICE L WALTON SCHOOL OF MEDICINE

Pharmacokinetics–3: Loading and Maintenance Dose

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Adapted from: Drug Concentration and Dosing

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Learning Objectives (2) Versions

After completing this brick, you will be able to:

1. Calculate loading and maintenance dose and explain the relevance of these adjustments in achieving steady state concentrations.
2. Explain key differences between loading and maintenance dose.

CASE CONNECTION

LC is a 54-year-old man who wants to stop the medications he has been taking long-term for his hypertension. "I feel like my heart goes fast and then slows down." "Is that normal?" he asks. Other than mild orthostasis and bradycardia from this hypertension, his exam is completely normal. However, his ECG confirms atrial fibrillation. The cardiologist adds Amiodarone (Class III Antiarrhythmic) to his drug list. The cardiologist counsels LC on Amiodarone and states that the initial oral dose will be 800 mg/day for six weeks. After six weeks, the cardiologist tells LC to stop taking the 800 mg/day tablets. A new prescription for 200 mg/day will be sent to the pharmacy for pick up.

What is the reason for lowering and changing the Amiodarone daily dose after six weeks? Consider your answer as you read, and we'll revisit LC at the end of the brick.

GO TO CONCLUSION ↓

How Do We Calculate Drug Dosage? (LO#1)

Now that we understand drug plasma concentration, volume of distribution (Vd), and steady-state concentration (C_{ss}), we can determine how much drug to administer. As mentioned, approximately 4-5 half-lives are needed to reach C_{ss} with regular dosing. If a drug's half-life is 1 day, 4-5 days are needed to reach a steady-state level.

Loading Dosing

What happens if a patient needs a therapeutic effect as soon as possible? This is where a loading dose comes in, which is a larger initial dose used to reach a target steady-state level much faster. The loading dose is then followed by a lower maintenance dose to keep the drug at C_{ss}. Also, if a drug has a large Vd then we need to know the loading dose so that we can achieve the effective minimum concentration.

Loading dose = $C_p \times V_d \div F$

where : C_p = target plasma concentration at steady state V_d = volume of distribution F = bioavailability (fraction of unchanged drug reaching systemic circulation)

Remember that F is equal to 1.0 when a drug is administered intravenously (IV) because the drug reaches the circulation immediately. For drugs administered orally, F is estimated at 0.7 because some metabolism occurs in the liver prior to drug circulation.

Sample Loading Dose Calculations. Using the fictional drug "fauxspirin," let's work through two examples.If the target concentration is 50 mg/mL, volume of distribution is 5 mL, and the drug is given IV, what would the loading dose of fauxspirin be (Figure 1)?

$$\text{Loading dose (mg)} = \frac{50\,\frac{\text{mg}}{\text{mL}} \times 5\ \text{mL}}{1} = 250\ \text{mg}$$

Figure 1

Now consider a case if the target concentration is 100 mg/mL and fauxspirin is administered orally with 50% bioavailability (Figure 2)?

$$\text{Loading dose (mg)} = \frac{100\,\frac{\text{mg}}{\text{mL}} \times 5\ \text{mL}}{0.5} = 1000\ \text{mg}$$

Figure 2

Maintenance Dosing

After administering a loading dose, the clinician must remember to switch to a lower maintenance dose to prevent drug toxicity. This is a preventable error that can unfortunately occur in the clinical setting.

CLINICAL CORRELATION

In a case study, an emergency department physician gave a seizing patient an IV phenytoin loading dose. However, the dose was not switched back to the appropriate maintenance dose, resulting in serum phenytoin levels three times greater than the maximum therapeutic level. The patient developed somnolence, severe ataxia (uncoordinated movements), and dysarthria (abnormal speech). Fortunately, no long-term harm occurred.

To calculate a maintenance dose (MD), remember that the goal is to maintain the concentration at use the following equation:

maintenance dose = $C_p \times CL \times \tau \div F$

where:

- C_p = target plasma concentration at steady state
- CL = clearance
- τ = dosage interval (time between doses), if not administered continuously
- F = bioavailability (fraction of unchanged drug reaching the systemic circulation)

To simplify, MD= (Dosing rate/F) X Dosing interval

In the example below (Figure 3), you can see that in order to reach Css, the drug had to be administered 5 times before a steady state concentration was achieved. On the other hand, when the loading dose was given, steady state was achieved with one administration of the loading dose.

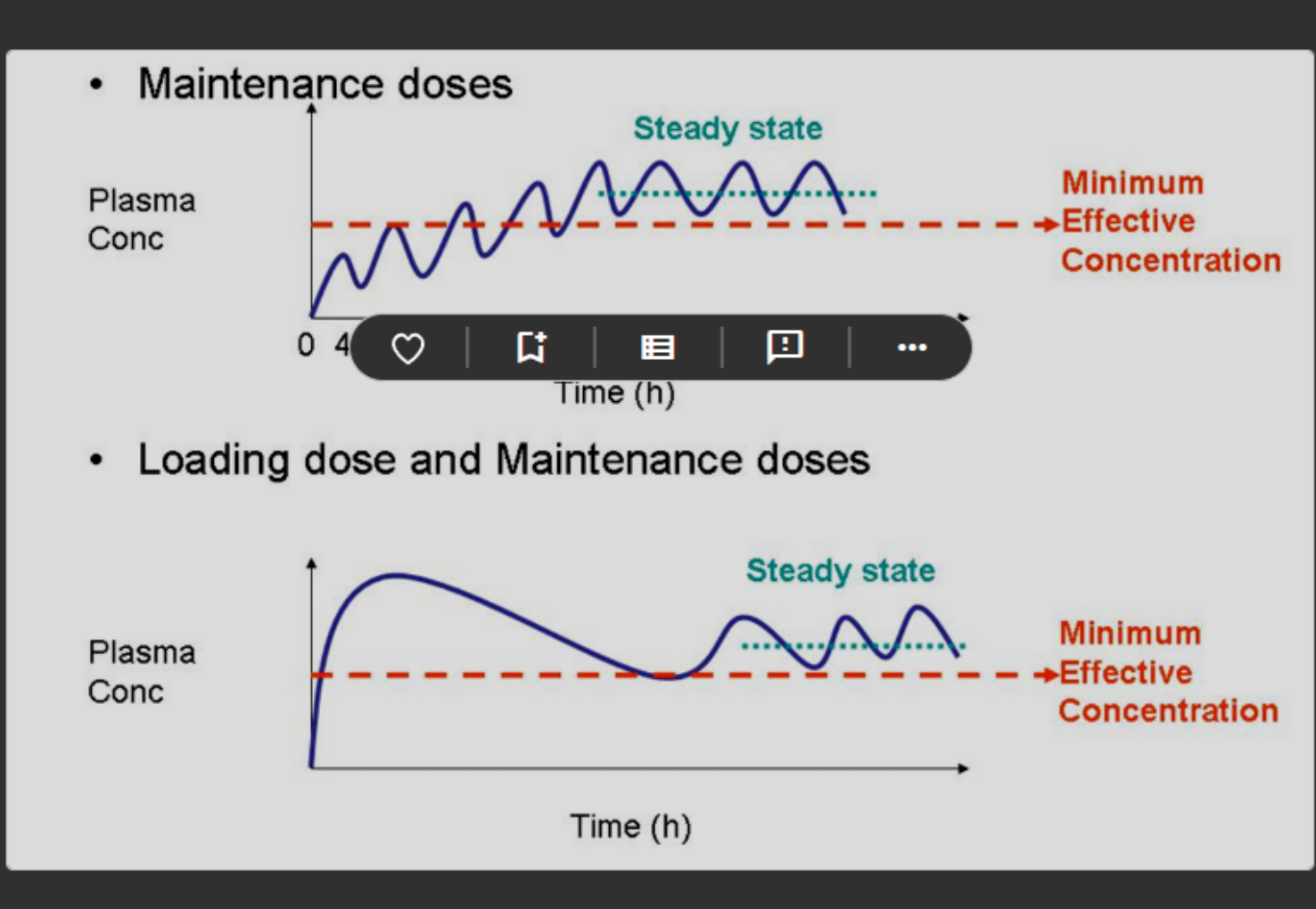

Figure 3

What are the key differences between Loading dose and Maintenance dose?

Lets review the key differences between the loading and maintenance dose (Table 1)

LOADING DOSE	MAINTENANCE DOSE
Dependent on V_d	Dependent on Clearance
Only few drug require Drug that are lipophilic E.g. Amiodarone, Digoxin	Most drugs need an MD
Rarely needs modifications	Should be modified based on patient or drug teristics

Table 1

Sample Maintenance Dose Calculations. Now, let's calculate the maintenance doses for the two fauxspirin examples. In the first, the target concentration was 50 mg/mL, and the drug was given through IV and continuously. What would the maintenance dose of fauxspirin be if clearance is 2 mL/h (Figure 4)?

$$\text{Maintenance dose} = \frac{50\,\frac{\text{mg}}{\text{mL}} \times 2\,\frac{\text{mL}}{\text{h}}}{1} = 100\ \text{mg/h}$$

Figure 4

What if the target concentration is 100 mg/mL, and fauxspirin is administered orally twice per day with 50% bioavailability (Figure 5)?

$$\text{Maintenance dose} = \frac{100\,\frac{\text{mg}}{\text{mL}} \times 2\,\frac{\text{mL}}{\text{h}} \times 12\text{h}}{0.5} = 4800\ \text{mg}$$

Figure 5

All that math is the basis for how to prescribe a drug and watch for its beneficial and toxic effects based on the prescription. That's pharmacotherapy!

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to LC, what was the reason to lower the dose for LC?

Amiodarone distributes slowly and widely across tissues such as fat, muscle, and highly perfused organs, necessitating long loading periods to speed up the drug's activity onset. Due to its high lipophilicity, oral amiodarone has a large volume of distribution (around 66 L/kg on average) and takes a long time to achieve stable plasma levels. Its absorption after oral administration is incomplete (about 30 to 70 percent) and varies significantly between individuals. Hence, a high-loading dose (800 mg) was prescribed for LC. A maintenance dose (200 mg) is established that helps achieve C_{ss} concentration for Amiodarone. In clinical practice, adverse effects of Amiodarone will also influence the maintenance dose for the patient.

Summary

- Drug dosage is calculated based on factors such as drug plasma concentration, volume of distribution, and steady-state concentration.
- A loading dose is a larger initial dose used to quickly reach a target steady-state level, followed by a lower maintenance dose to maintain the steady-state level.
- Maintenance dos vent drug toxicity, taking into account factors such as target plasma concentration, clearance, dosing interval, and bioavailability.
- Doubling the infusion rate of a drug without changing the infusion time will double the steady-state concentration.

Review Questions

1. A patient was started on daily oral digoxin for the management of atrial fibrillation. F=0.6, Vd=500 L, and Cl= 120 L/day. What is the oral loading dose of digoxin is appropriate to rapidly achieve a peak concentration of 1.5 mcg/L?

A. ○ 1250 mg
B. ● 1.25 mg
C. ○ 1.50 mg
D. ○ 1.0 g
E. ○ 2.25 mg

Hide Explanation

The correct answer is 1.25 mg (B). The answer 1250 mg has incorrect unit. Answers (C), (D), and (E) are incorrect calculation.

2. If the steady-state concentration of drug X is 40 mg/mL, and the drug is being cleared at 20 mL/h, which of the following is the rate of drug (in mg/h) administered to the patient?

A. ○ 2 mg/h
B. ○ 4 mg/h
C. ○ 50 mg/h
D. ○ 200 mg/h
E. ● 800 mg/h

Hide Explanation

The correct answer is 800 mg/h (E). Steady-state concentration (mg) is calculated by dividing rate of drug intake (mg/h) by drug clearance (mL/h). Thus, if you already know the steady-state concentration, you can multiply by drug clearance to get rate of drug intake (40 mg/mL × 20 mL/h = 800 mL/h). The other values are incorrect: 2 mg/h (A) divides steady-state concentration by drug cleared (which does not yield rate of drug intake), 4 mg/h (B) and 50 mg/h (C) are drug intake rates too small to reach the desired steady-state concentration, and 200 mg/h (D) is only a quarter of the necessary drug intake rate to reach the desired steady-state concentration.

3. What is the primary benefit of administering a loading dose?

A. ○ Gradual dose escalation
B. ● Rapid achievement of C_{ss}
C. ○ Reduced side effects
D. ○ Reduced frequency of dosing

Hide Explanation

The correct answer is rapid achievement of Css (B). Administering a loading dose allows the drug to reach steady state concentration quickly. Gradual dose escalation (A) does not achieve this. Reduced side effects (C) are not primary benefits of a loading dose. Reduced frequency of dosing (D) is also not a primary benefit.

How would you rate this Brick?

👍 👎

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Calcul ain the relevance of these adjustments in achieving steady state concentrations.~~

Hide Explanation

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01 ☑ ~~Calculate loading and maintenance dose and explain the relevance of these adjustments in achieving steady state concentrations.~~

02 ☑ ~~Explain key differences between loading and maintenance dose.~~

Objective Confidence: 2/2